

Stat 707 HW1

Alex Caramanico

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Stan Code

```
library(rstan)

## Warning: package 'rstan' was built under R version 4.2.3

## Loading required package: StanHeaders

## Warning: package 'StanHeaders' was built under R version 4.2.3

##
## rstan version 2.32.5 (Stan version 2.32.2)

## For execution on a local, multicore CPU with excess RAM we recommend calling
## options(mc.cores = parallel::detectCores()).
## To avoid recompilation of unchanged Stan programs, we recommend calling
## rstan_options(auto_write = TRUE)
## For within-chain threading using 'reduce_sum()' or 'map_rect()' Stan functions,
## change 'threads_per_chain' option:
## rstan_options(threads_per_chain = 1)

## Do not specify '-march=native' in 'LOCAL_CPPFLAGS' or a Makevars file

hw1.3a.code= 'data {int<lower = 0> n;
                array[n] int y;}

parameters {real<lower=0> theta;}

model {
  y ~ binomial(10, theta);
  theta ~ beta(1, 1);}'

hw1.3b.code= 'data {int<lower = 0> n;
                array[n] int y;}

parameters {real theta;}
```

```

model {
  y ~ binomial(10, theta);
  theta ~ beta(0.5, 0.5);}

hw1.3c.code= 'data {int<lower = 0> n;
               array[n] int y;}

parameters {real theta;}

model {
  y ~ binomial(10, theta);
  theta ~ beta(0.1, 0.1);}

```

3a Samples

```

hw.3a.model = stan_model(model_code=hw1.3a.code,model_name='HW1a')
hw1.3a.draws = sampling(
  hw.3a.model,
  data = list(y = c(2,6,2,5,7),n=5),
  chains = 2,
  cores = 2,
  iter = 10000,
  warmup = 5000,
  seed = 303
)

```

Display Results

```
print(hw1.3a.draws)
```

```

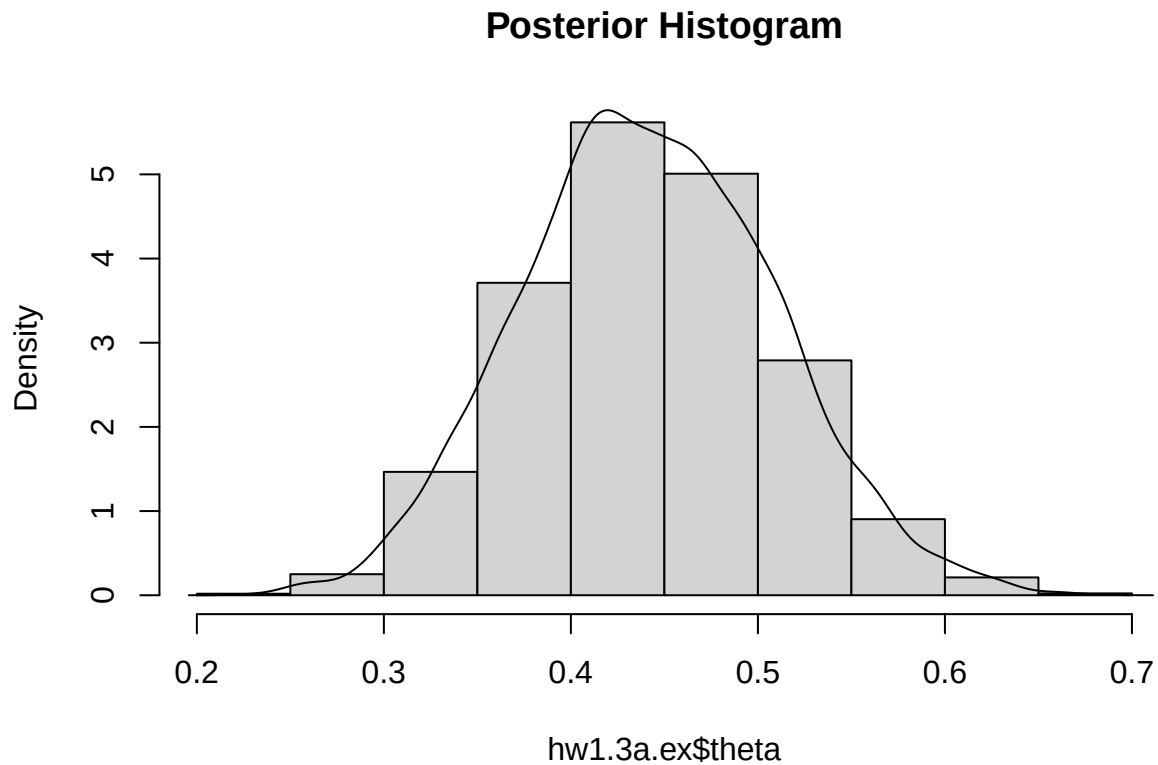
## Inference for Stan model: HW1a.
## 2 chains, each with iter=10000; warmup=5000; thin=1;
## post-warmup draws per chain=5000, total post-warmup draws=10000.
##
##          mean se_mean   sd  2.5%   25%   50%   75%  97.5% n_eff Rhat
## theta    0.44    0.00 0.07   0.31   0.40   0.44   0.49   0.58  3117   1
## lp__   -35.60    0.01 0.69  -37.59 -35.76 -35.33 -35.16 -35.11  3700   1
##
## Samples were drawn using NUTS(diag_e) at Tue Feb 13 16:08:47 2024.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).

```

```

hw1.3a.ex = extract(hw1.3a.draws)
# For prior Beta(1,1): posterior mean of theta = 0.44,
#   and posterior standard deviation of theta = 0.07
hist(hw1.3a.ex$theta,main='Posterior Histogram',prob=TRUE)
lines(density(hw1.3a.ex$theta))

```



3b Samples

```
hw.3b.model = stan_model(model_code=hw1.3b.code,model_name='HW1b')
hw1.3b.draws = sampling(
  hw.3b.model,
  data = list(y = c(2,6,2,5,7),n=5),
  chains = 2,
  cores = 2,
  iter = 10000,
  warmup = 5000,
  seed = 303
)
```

Display Results

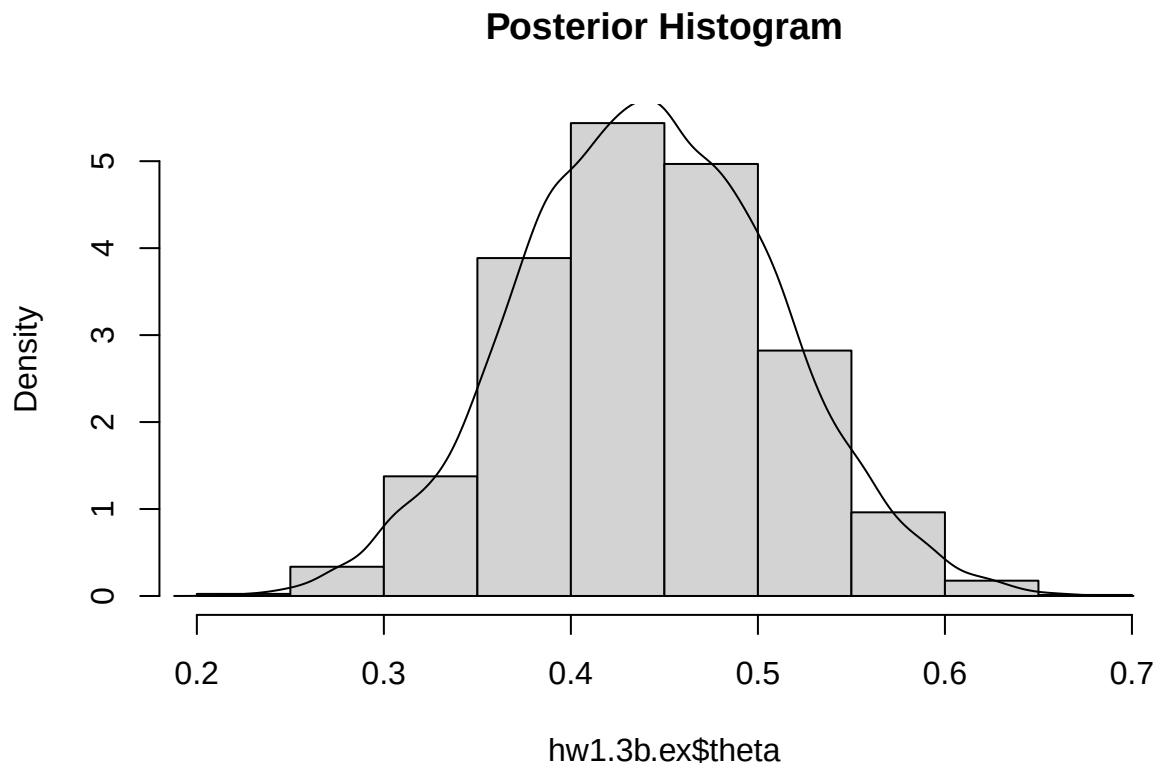
```
print(hw1.3b.draws)
```

```
## Inference for Stan model: HW1b.
## 2 chains, each with iter=10000; warmup=5000; thin=1;
## post-warmup draws per chain=5000, total post-warmup draws=10000.
```

```
##
##          mean se_mean  sd   2.5%   25%   50%   75%  97.5% n_eff Rhat
## theta    0.44    0.00 0.07   0.31   0.39   0.44   0.49   0.58 3755   1
## lp__    -34.07    0.01 0.66  -35.96 -34.21 -33.82 -33.65 -33.60 3424   1
##
## Samples were drawn using NUTS(diag_e) at Tue Feb 13 16:09:23 2024.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).
```

```
hw1.3b.ex = extract(hw1.3b.draws)
# For prior Beta(0.5,0.5): posterior mean of theta = 0.44,
#   and posterior standard deviation of theta = 0.07

hist(hw1.3b.ex$theta,main='Posterior Histogram',prob=TRUE)
lines(density(hw1.3b.ex$theta))
```



3c Samples

```
hw.3c.model = stan_model(model_code=hw1.3c.code,model_name='HW1c')
hw1.3c.draws = sampling(
  hw.3c.model,
  data = list(y = c(2,6,2,5,7),n=5),
```

```

chains = 2,
cores = 2,
iter = 10000,
warmup = 5000,
seed = 303
)

```

Display Results

```
print(hw1.3c.draws)
```

```

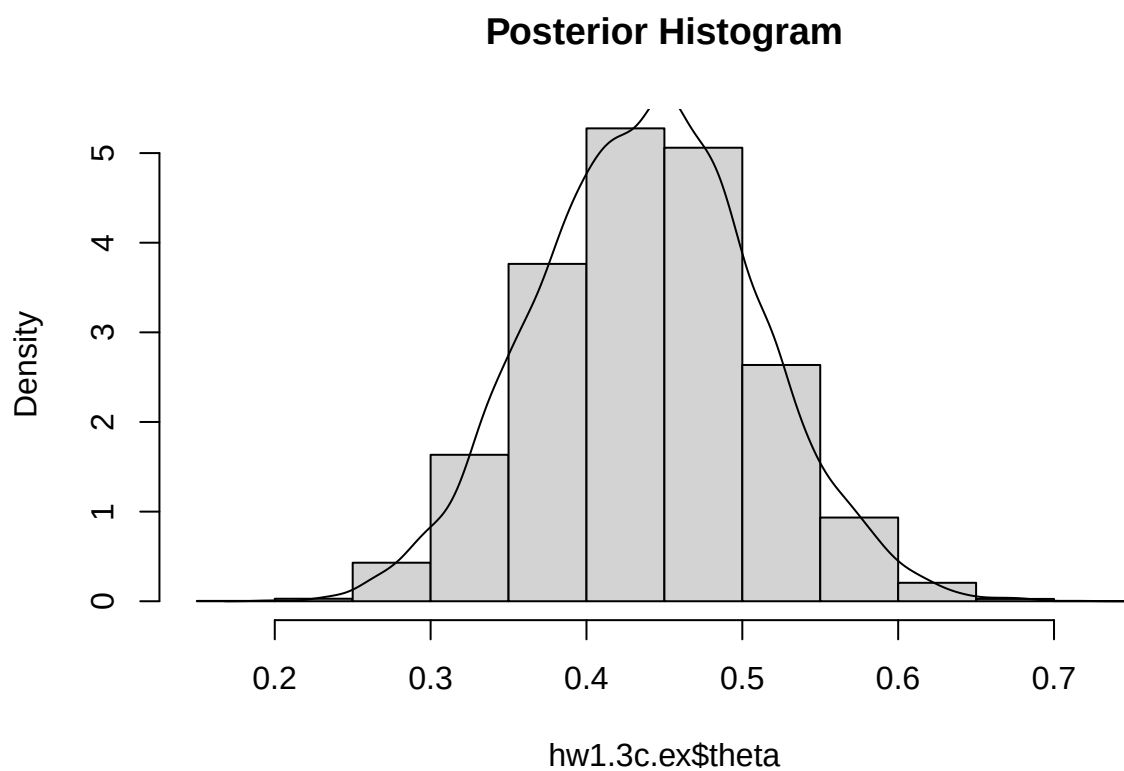
## Inference for Stan model: HW1c.
## 2 chains, each with iter=10000; warmup=5000; thin=1;
## post-warmup draws per chain=5000, total post-warmup draws=10000.
##
##          mean se_mean   sd  2.5%  25%   50%   75%  97.5% n_eff Rhat
## theta    0.44    0.00 0.07   0.30  0.39   0.44   0.49   0.58 3221   1
## lp__   -33.53    0.01 0.70 -35.53 -33.69 -33.26 -33.09 -33.04 4118   1
##
## Samples were drawn using NUTS(diag_e) at Tue Feb 13 16:09:59 2024.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).

```

```

hw1.3c.ex = extract(hw1.3c.draws)
# For prior Beta(0.1,0.1): posterior mean of theta = 0.44,
#   and posterior standard deviation of theta = 0.07
hist(hw1.3c.ex$theta,main='Posterior Histogram',prob=TRUE)
lines(density(hw1.3c.ex$theta))

```



Ala Coramano Stat 707 HW1 2-13-24

$$\hat{\theta}_{MLE} = \frac{Y}{n}$$

$$\hat{\theta}_{Boris} = \frac{Y+1}{n+2}$$

$$MSE = \frac{\theta(1-\theta)}{n}$$

→ find MSE, when would we prefer this

$$MSE = V(\hat{\theta}) + [B(\hat{\theta})]^2 = E[\hat{\theta} - E(\hat{\theta})]^2 + [E(\hat{\theta}) - \theta]^2$$

$$a) V(\hat{\theta}) = V\left(\frac{Y}{n}\right) = \frac{1}{n^2} \text{Var}(Y) = \frac{\text{Var}(Y)}{n^2} = \frac{n\theta(1-\theta)}{n^2}$$

$$b) B(\hat{\theta})^2 = (E(\hat{\theta}) - \theta)^2 = \left(\frac{n+1}{n+2} - \theta\right)^2 = \left(\frac{n\theta+1}{n+2} - \theta\right)^2 = \left(\frac{1-2\theta}{n+2}\right)^2 = \frac{1-4\theta+4\theta^2}{(n+2)^2}$$

$$\text{then } MSE = \frac{n\theta(1-\theta) + 1-4\theta+4\theta^2}{(n+2)^2} = \frac{n\theta - n\theta^2 + 1 - 4\theta + 4\theta^2}{(n+2)^2} = \boxed{\frac{\theta^2(4-n) + \theta(n-4) + 1}{(n+2)^2}}$$

Discussion: $V(\hat{\theta}_{MLE}) = \text{Var}\left(\frac{Y}{n}\right) = \frac{1}{n^2} \text{Var}(Y) = \frac{n\theta(1-\theta)}{n^2}$

$$V(\hat{\theta}_{Boris}) = \text{Var}\left(\frac{Y+1}{n+2}\right) = \frac{1}{(n+2)^2} \text{Var}(Y+1) = \frac{n\theta(1-\theta)}{(n+2)^2}$$

The variance of the Boris estimator is smaller, which may be preferable (although it is biased) because it gives more consistent (less variance) estimates.

The MSE of both estimators goes to 0 as $n \rightarrow \infty$

If we consider a small sample size, say $n=5$, then

$$V(\hat{\theta}_{MLE}) = \frac{5\theta(1-\theta)}{25}, \quad V(\hat{\theta}_{Boris}) = \frac{5\theta(1-\theta)}{49}$$

if $n=1000$,

$$V(\hat{\theta}_{MLE}) = \frac{1000\theta(1-\theta)}{1000^2}, \quad V(\hat{\theta}_{Boris}) = \frac{1000\theta(1-\theta)}{1002^2}$$

It can be seen that the variance of the Bayesian estimator is much lower than the MLE with smaller sample sizes versus large ones. The same thing applies to the MSE, and lower MSE is inherently good because there are literally smaller errors.

In general, it is a question of controlling bias vs. controlling variance

2. Let $Y|θ \sim \text{Bin}(n, θ)$, n known, $θ$ unknown (success)

Assume a $\text{Beta}(\alpha, \beta)$ prior on $θ$.

a) find the posterior distribution of $θ$.

$$P(θ|y) = \frac{P(y|θ) \pi(θ)}{\int P(y|θ) \pi(θ) dθ}$$

$$\pi(θ) = \frac{θ^{\alpha-1} (1-θ)^{\beta-1}}{B(\alpha, \beta)}$$

$$P(y|θ) = \prod_{i=1}^n \binom{n}{y_i} θ^{y_i} (1-θ)^{n-y_i}$$

$$P(y|θ) \pi(θ) = θ^{\sum y_i} (1-θ)^{n-\sum y_i} θ^{\alpha-1} (1-θ)^{\beta-1} \cdot \prod_{i=1}^n \binom{n}{y_i}$$

$$= C(y) θ^{\sum y_i + \alpha - 1} (1-θ)^{n - \sum y_i + \beta - 1} \cdot \frac{\prod_{i=1}^n \binom{n}{y_i}}{B(\alpha, \beta)}$$

C(y) no theta

$$P(θ|y) = \frac{C(y) θ^{\sum y_i + \alpha - 1} (1-θ)^{n - \sum y_i + \beta - 1}}{C(y) \int_0^1 θ^{\sum y_i + \alpha - 1} (1-θ)^{n - \sum y_i + \beta - 1} dθ}$$

$$\propto θ^{(\alpha + \sum y_i - 1)} (1-θ)^{(\beta + n - \sum y_i - 1)}$$

$$\text{or } \text{Beta}(\alpha + \sum y_i, \beta + n - \sum y_i)$$

$$\rightarrow = \frac{1}{B(\alpha + \sum y_i, \beta + n - \sum y_i)} θ^{\sum y_i + \alpha - 1} (1-θ)^{n - \sum y_i + \beta - 1}$$

Are they $\sum y_i$ or just y_i ?

b) Find the mean and standard deviation of the posterior distribution

Expectation value of Beta is $\frac{\alpha}{\alpha + \beta} \rightarrow \frac{\alpha + \sum y_i}{\alpha + \sum y_i + \beta + n - \sum y_i} = \frac{\alpha + \sum y_i}{\alpha + \beta + n}$

Var of beta is $\frac{\alpha \beta}{(\alpha + \beta)^2 (\alpha + \beta + 1)} \rightarrow \frac{(\alpha + \sum y_i) (\beta + n - \sum y_i)}{(\alpha + \beta + n)^2 (\alpha + \beta + n + 1)}$ * may be simplified?